

Homework I - Task 4:

Word-in-Context (WiC) Disambiguation

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1 Introduction

Word-in-Context (WiC) Disambiguation [2] is a binary NLP classification task. Given a word w occurring in two distinct sentences $s1$ and $s2$, the objective of the task is to analyze the context of the two sentences and determine if the meaning of the word presents **ambiguity** or not:

- 0 if the word w has *not* the same meaning in the two sentences $s1$ and $s2$;
- 1 if the word w has the same meaning in the two sentences $s1$ and $s2$.

2 Dataset Format

From the provided EVALITA dataset, only the files related to the binary task should be used.

Specifically, the 'train.json', 'dev.json' and 'test.json' files are extracted from binary folder to execute the task. An example of the files format is provided in Listing 1 to enhance comprehension.

3 Methodology

3.1 Data Reformatting

To evaluate the linguistic skills of Large Language Models (LLMs), the data is converted into JSONL standard. An example of this format is provided in Listing 2 to enhance comprehension.

The most important addition concerns a list of strings representing the possible **choices** available to the model, namely "*Ambiguo*" and "*Non ambiguo*", indicating the meaning nature of the given word in the two sentences.

3.2 Prompts Formulation

The primary goal of this work is to formulate prompts able to improve the performance and manage the behavior of LLMs. In Listing 3, the list of suitable prompts is provided, saved in JSONL standard.

3.3 Models

To evaluate the quality of the prompts, the **mDeBERTa** model [1] was chosen. It is a *multilingual model*, pretrained on over 100 languages (including Italian), and represents a common solution for text classification tasks.

3.4 Prompts Evaluation

Accuracy, precision, recall and F1 metrics are computed to evaluate the prompts. More attention is given to **accuracy score** in relation to **F1-score**: accuracy score alone does not provide complete information about performance, whereas F1-score represents the harmonic mean between precision and recall scores.

The Train and Test Set **confusion matrices** in Figure 1 e Figure 2 graphically summarize the overall performance of the selected model in Train and Test Sets.

(**Note:** Dev Set confusion matrices and other statistics can be observed in the Python script.)

4 Final Considerations

The results displayed in Table 1 highlight that the model has achieved good performance without encountering any execution time issues.

Additionally, **Prompt 1** and **Prompt 3** appear to be slightly better formulated and clearer than Prompt 2. The scores are not very high because a pretrained model was used. By employing *fine-tuning*, the overall results could potentially increase.

5 Code Instructions

The Python script was run on Google Colab and the dataset was uploaded on Google Drive to make it accessible in the notebook. GPU is necessary to run models and avoid execution time problems.

6 Tables and Figures

Listing 3: Prompts list

Listing 1: Example of a Dev Set sample

```
{ "id": "delicatezza.noun.14",  
  "lemma": "delicatezza",  
  'sentence1': 'Il presidente  
AZZOLLINI , stante la delicatezza  
dei temi oggetto del  
provvedimento , ritiene opportuno  
richiedere la predisposizione di  
una relazione tecnica .',  
  'sentence2': 'Montare la panna  
senza zucchero in modo che  
diventi molto solida , quindi  
unirla al composto avendo cura di  
mescolarla con molta  
delicatezza .',  
  "start1": 36,  
  "end1": 47,  
  "start2": 129,  
  "end2": 140,  
  "label": 0 }
```

Listing 2: Exaple of a Dev Set reformatting sample

```
{ 'id': 0,  
  'lemma': 'delicatezza',  
  'sentence1': 'Il presidente  
AZZOLLINI , stante la delicatezza  
dei temi oggetto del  
provvedimento , ritiene opportuno  
richiedere la predisposizione di  
una relazione tecnica .',  
  'sentence2': 'Montare la panna  
senza zucchero in modo che  
diventi molto solida , quindi  
unirla al composto avendo cura di  
mescolarla con molta  
delicatezza .',  
  'start1': 36,  
  'end1': 47,  
  'start2': 129,  
  'end2': 140,  
  'choices': [ 'Ambiguo',  
               'Non ambiguo'],  
  'label': 0 }
```

```
{ 'prompt': "Il significato della  
parola '{lemma}' risulta  
(Ambiguo) o (Non ambiguo)  
nella frase '{sentence1}'  
e nella frase '{sentence2}'?" }
```

```
{ 'prompt': "Confronta le frasi  
'{sentence1}' e '{sentence2}'  
per determinare se il significato  
della parola '{lemma}'  
\u00e8 (Ambiguo) o (Non ambiguo)" }
```

```
{ 'prompt': "Valuta se il  
significato della parola  
'{lemma}' risulta essere  
(Ambiguo) o (Non Ambiguo)  
nella frase '{sentence1}'  
e nella frase '{sentence2}'." }
```

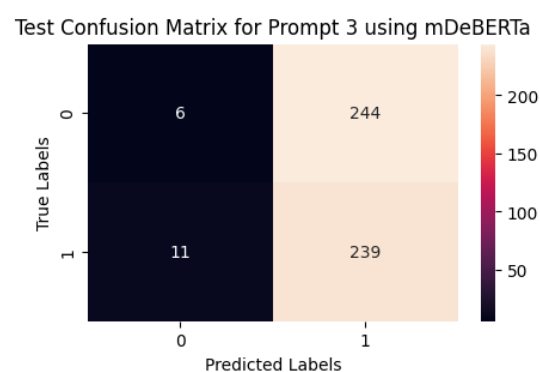
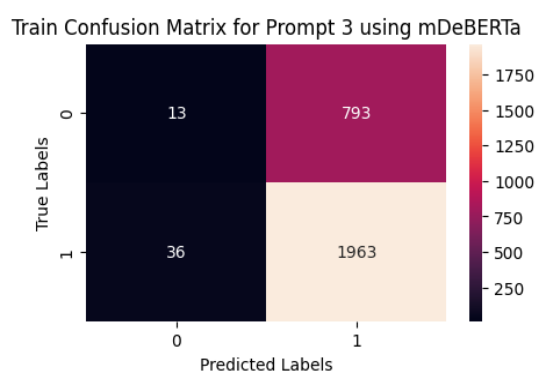
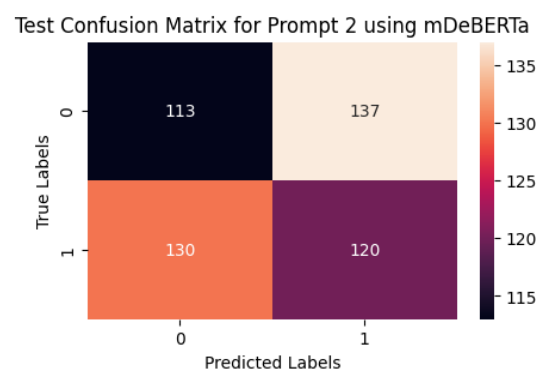
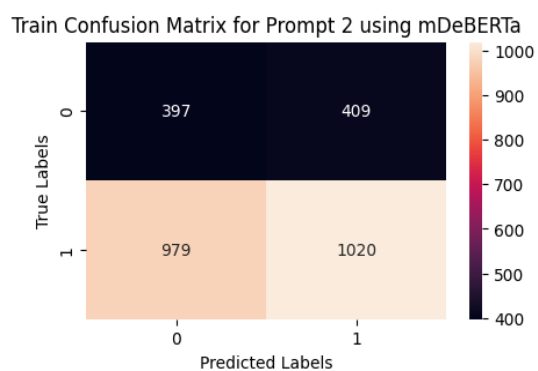
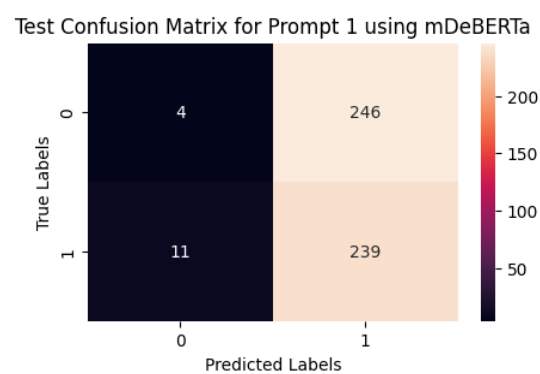
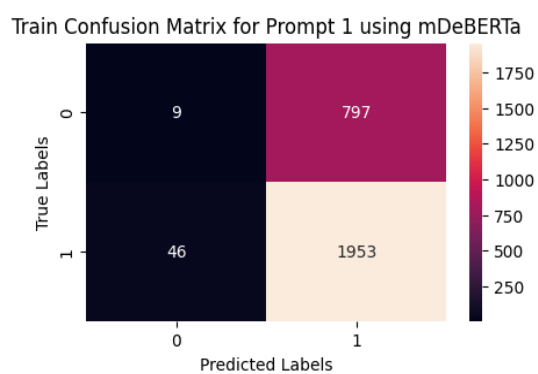


Figure 1: Train Confusion Matrices using mDeBERTa

Figure 2: Test Confusion Matrices using mDeBERTa

	Prompt 1	Prompt 2	Prompt 3
Train Accuracy mDeBERTa Base Multilingual	0.699	0.505	0.704
Train F1-score mDeBERTa Base Multilingual	0.822	0.595	0.826
Test Accuracy mDeBERTa Base Multilingual	0.486	0.466	0.490
Test F1-score mDeBERTa Base Multilingual	0.650	0.473	0.652
Dev Accuracy mDeBERTa Base Multilingual	0.486	0.550	0.488
Dev F1-score mDeBERTa Base Multilingual	0.651	0.563	0.656

Table 1: Performance Comparison Table

References

- [1] Moritz Laurer. mdeberta-v3-base-xnlimultilingual-nli-2mil7. [Hugging Face](#).
- [2] Mohammad Taher Pilehvar and Jose Camacho-Collado. *WiC: the Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations*. Association for Computational Linguistics, 2019.